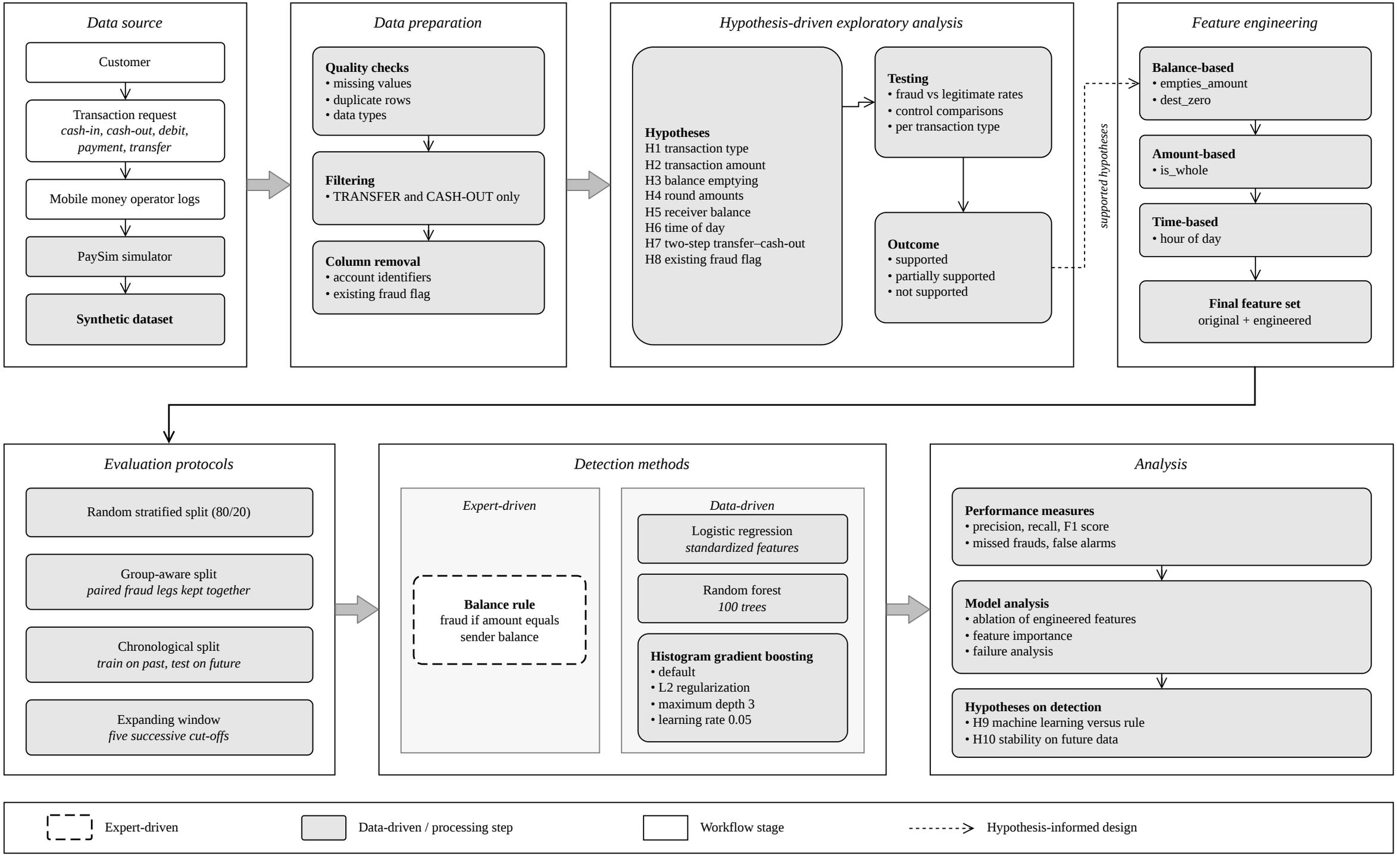

Data source
Customer
Transaction request
cash-in, cash-out, debit, payment, transfer
Mobile money operator logs
PaySim simulator
Synthetic dataset
Data preparation
Quality checks
• missing values
• duplicate rows
• data types
Filtering
• TRANSFER and CASH-OUT only
Column removal
• account identifiers
• existing fraud flag
Hypothesis-driven exploratory analysis
Hypotheses
H1 transaction type
H2 transaction amount
H3 balance emptying
H4 round amounts
H5 receiver balance
H6 time of day
H7 two-step transfer–cash-out
H8 existing fraud flag
Testing
• fraud vs legitimate rates
• control comparisons
• per transaction type
Outcome
• supported
• partially supported
• not supported
supported hypotheses
Feature engineering
Balance-based
• empties_amount
• dest_zero
Amount-based
• is_whole
Time-based
• hour of day
Final feature set
original + engineered
Evaluation protocols
Random stratified split (80/20)
Group-aware split
paired fraud legs kept together
Chronological split
train on past, test on future
Expanding window
five successive cut-offs
Detection methods
Expert-driven
Balance rule
fraud if amount equals sender balance
Data-driven
Logistic regression
standardized features
Random forest
100 trees
Histogram gradient boosting
• default
• L2 regularization
• maximum depth 3
• learning rate 0.05
Analysis
Performance measures
• precision, recall, F1 score
• missed frauds, false alarms
Model analysis
• ablation of engineered features
• feature importance
• failure analysis
Hypotheses on detection
• H9 machine learning versus rule
• H10 stability on future data
Expert-driven
Data-driven / processing step
Workflow stage
Hypothesis-informed design